

LAB SESSION CORRELATION REG

Import the libraries and load the dataset.

Select the columns to be correlated in two variables.

```
x= df[“col1”], y = df[“col2”]
```

Calculate Correlation

```
correlation = x.corr(y)
print("Correlation:", correlation)
```

Calculate Covariance

```
covariance = np.cov(x, y)
print("Covariance Matrix:\n", covariance)
```

Visualize relationship (Scatter plot)

```
sns.scatterplot(x=x, y=y)
plt.title("Relationship between Sepal Length and Petal Length")
plt.show()
```

Detect Outliers

```
Q1= df[“col1”].quantile (0.25)
```

```
Q3 = df[“col1”].quantile (0.75)
```

```
IQR = Q3 - Q1

lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR

print(lower, upper)
```

#Remove outliers

```
clean_data = data[(data['sepal_length'] >= lower) &
                  (data['sepal_length'] <= upper)]

print("Original shape:", data.shape)
print("After removing outliers:", clean_data.shape)
```

Recalculate correlation after removing outliers

```
new_corr = clean_data['sepal_length'].corr(clean_data['petal_length'])
print("Correlation after removing outliers:", new_corr)
```

Interpretation of results

Correlation (r) → Measures strength & direction of relationship

- Range: -1 to +1
- Close to +1 → strong positive relation

Covariance → Measures how two variables vary together

- Positive → move together
- Negative → move oppositely

To draw Histogram

```
sns.histplot(data, kde=True)
```

OR

```
import matplotlib.pyplot as plt
```

```
plt.hist(data)
```

```
plt.show()
```

```
plt.title("Histogram with KDE - Customer Spending")
```

```
plt.xlabel("Spending (₹)")
```

```
plt.ylabel("Frequency")
```

```
plt.show()
```

Question:

Import the important libraries and download the 'E-Commerce_Cust_data'

Q1 Find the covariance and correlation between Items Purchased and Total Spend and interpret the result

Q2 Draw a scatter plot for the above two fields

Q3 Draw the histogram of the 'Membership' column of the dataset

Q4 Draw the histogram of the 'Satisfaction level' column of the dataset

Q5 Draw a scatter plot for Total spend and average rating

Q5 Create a data frame for values $X = [10, 20, 30, 40, 1000]$ and $Y = [15, 25, 35, 45, 1100]$

- a) Calculate covariance and correlation between X and Y
- b) Check for outliers in X and Y and remove them
- c) Recalculate the covariance and correlation between X and Y after removing outliers.

